Technical Analysis using R

Visualization Phase

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Portfolio dashboard

An interactive R Shiny dashboard for AAPL, MSFT, GOOGL, AMZN and TSLA. The controls select dates, daily, weekly or monthly bars, and line, candlestick or area charts. Moving averages overlay price, with RSI and MACD on separate date panels. SMA crossings mark Buy or Sell on the chart, and the table shows Hold on other bars.

Source code and data

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Install shiny, ggplot2 and quantmod, then open app.R in RStudio and click Run App. The app opens with saved Yahoo Finance data from 2025. Select the live source and click Load data to fetch another period.

Verification

The R checks passed for all three chart types, aggregation, indicators and crossover signals. The Shiny server rendered daily candlesticks and monthly area charts using saved historical data. A live Yahoo Finance request was not verified.

AI assistance: ChatGPT (Codex) helped adapt the code, run tests and prepare documentation.